MANY-ELECTRON ATOMS

General Chemistry I, Lecture Series 7

Qixi Mi

Reading:

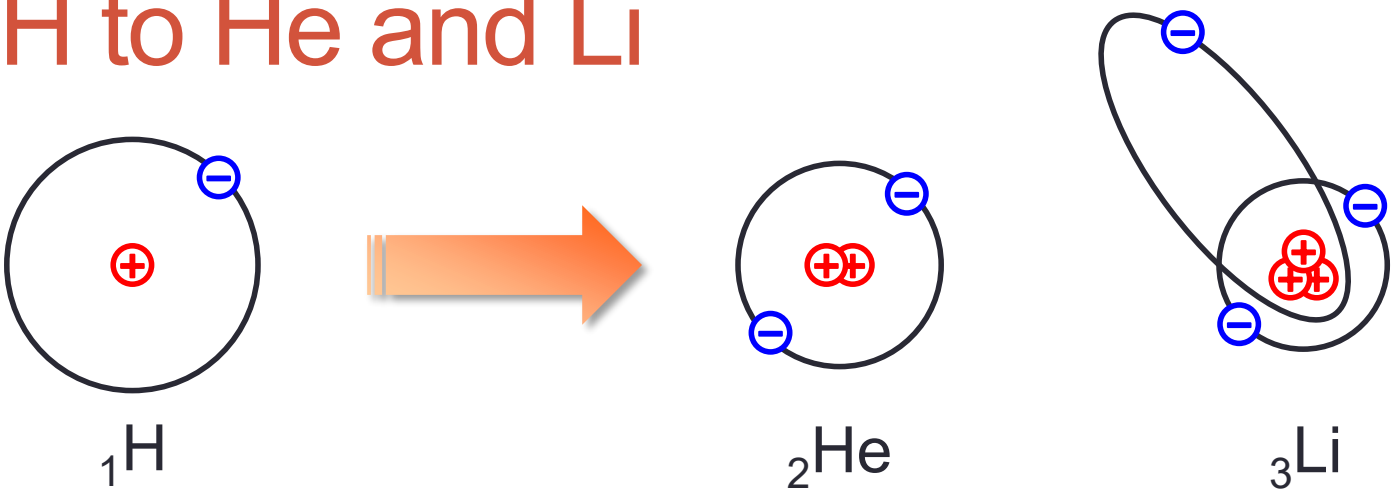
OGB8 §5.2, §5.3

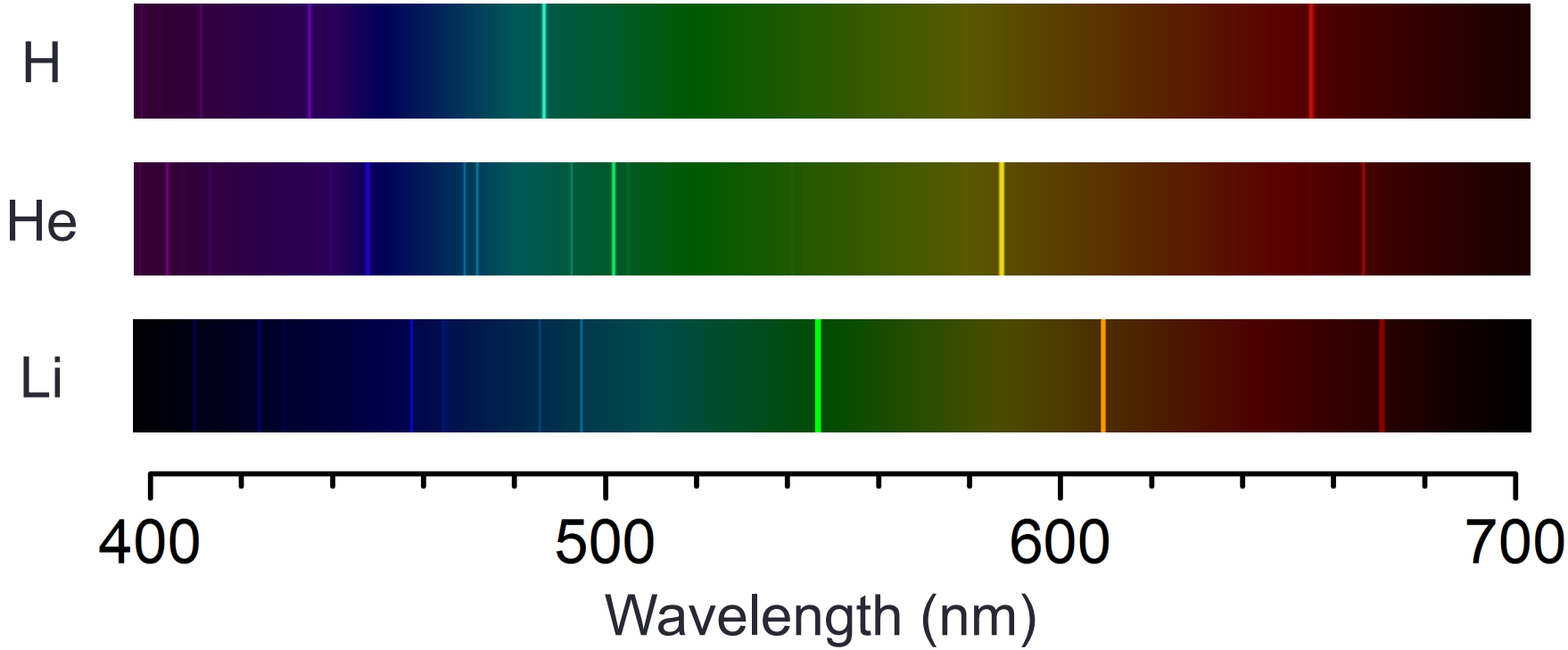
© Qixi Mi 2014–25, except pictures from the Internet.

Outline

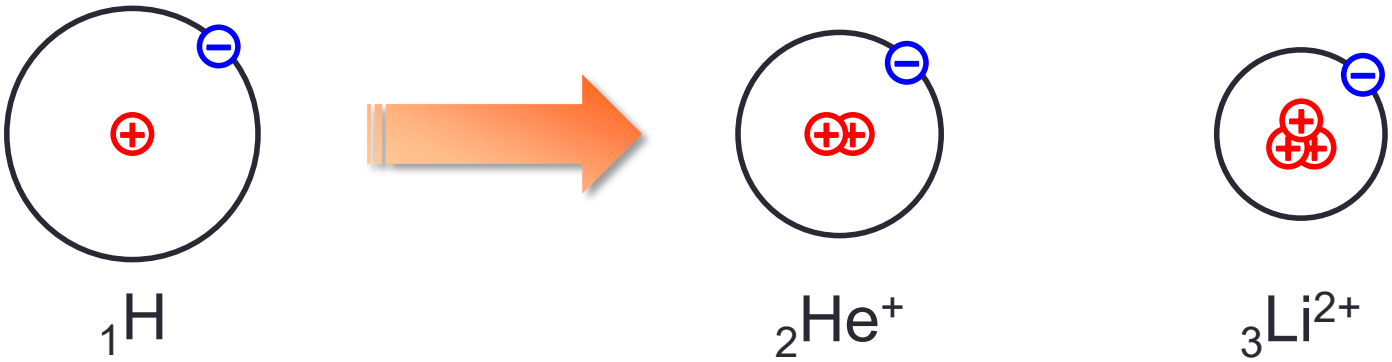
- Interactions in many-electron atoms
- Repulsion between electrons
- Aufbau principle
- Electron configurations

From H to He and Li





Hydrogen-Like Ions

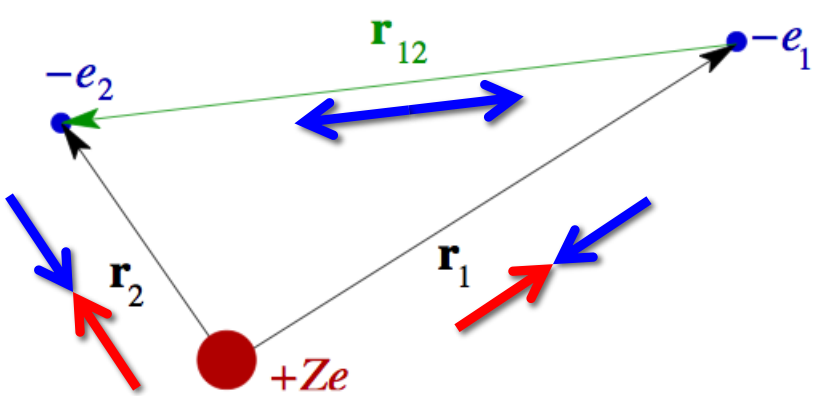


Coulomb potential V	$-\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$	$-\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$
1s orbital $\psi_{1s} \propto$	$\exp(-r/a_0)$	$\exp(-Zr/a_0)$
Orbital radius r_n	n^2a_0	$\frac{n^2}{Z}a_0$
Orbital energy E_n	$-\frac{\mathrm{Ry}}{n^2}$	$-Z^2\frac{\mathrm{Ry}}{n^2}$

a_0 = 53 pm
Ry = 13.6 eV

Single- vs. Many-Electron Atoms

	H, He^+, Li^{2+}	He, Li, …
Interactions	*Z*-*e* attraction	*Z*-*e* attraction *e*-*e* repulsion electron spins
Orbitals	*e* in a single orbital	*e* in multiple orbitals
Coordinates	*Z* at *O*, *e* at *r*	*Z* at *O*, e_1 at r_1, e_2 at r_2, …
Schrödinger Eqn.	Analytical solutions	Approximate solutions



An electron is attracted by $+Z$ but repelled by $-(Z-1)e$

Chemistry and Mathematics

Paul A. M. Dirac
(1902–1984)

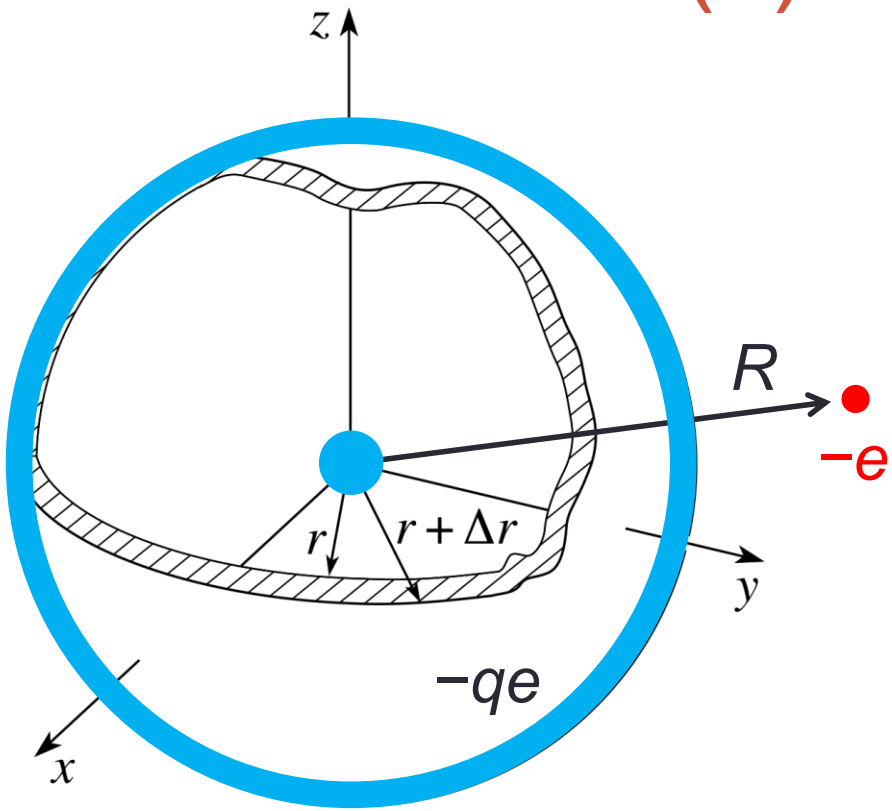
The fundamental laws necessary for the mathematical treatment of a large part of physics and the whole of chemistry are thus completely known, and the difficulty lies only in the fact that application of these laws leads to equations that are too complex to be solved.

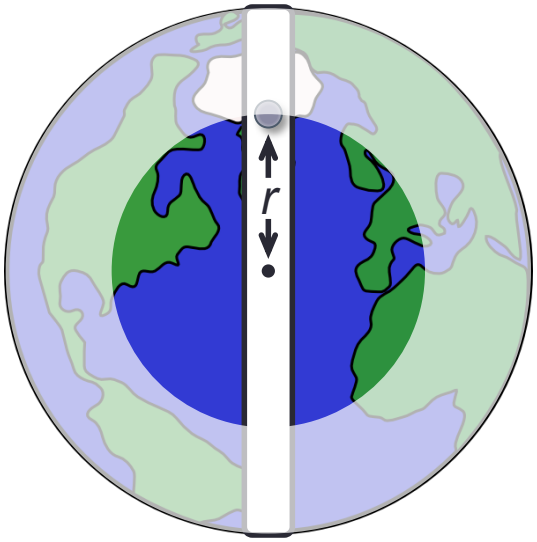
The many-electron Schrödinger Equation can only be solved approximately.

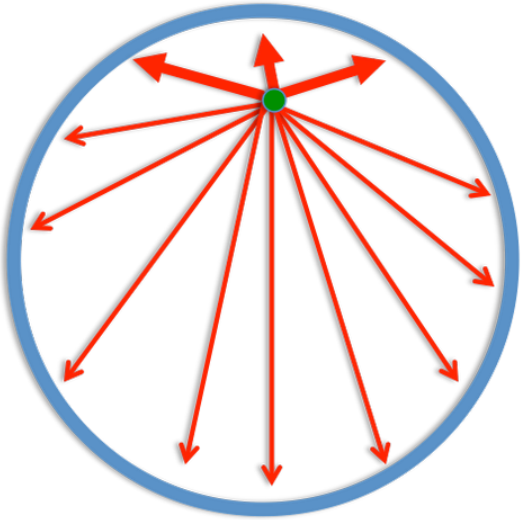
Outline

- Interactions in many-electron atoms
- Repulsion between electrons
  - Electrons in different shells
  - Electrons in the same shell
  - Electrons in the same orbital
- Aufbau principle
- Electron configurations

The Shell Model (1)



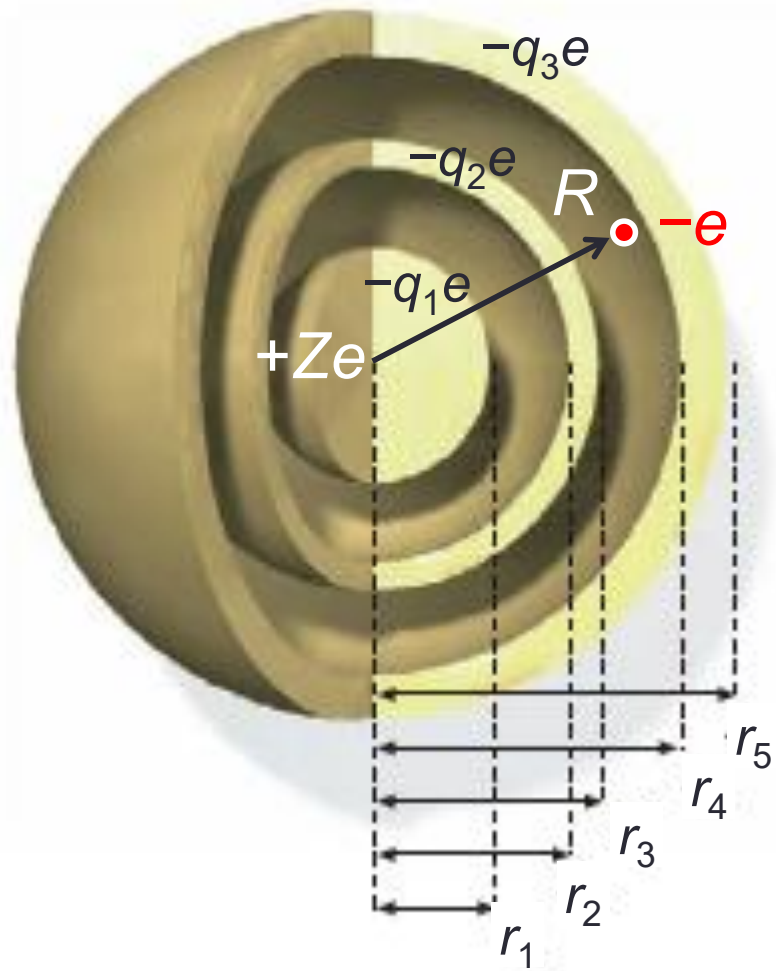




$$F = \frac{1}{4\pi\varepsilon_0}\frac{qe^2}{R^2} \text{ when } R \geq r$$

$$F = 0 \text{ when } R < r$$

The Shell Model (2)



$$F = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{R^2} + \frac{1}{4\pi\varepsilon_0}\frac{q_1e^2}{R^2} + \frac{1}{4\pi\varepsilon_0}\frac{q_2e^2}{R^2}$$

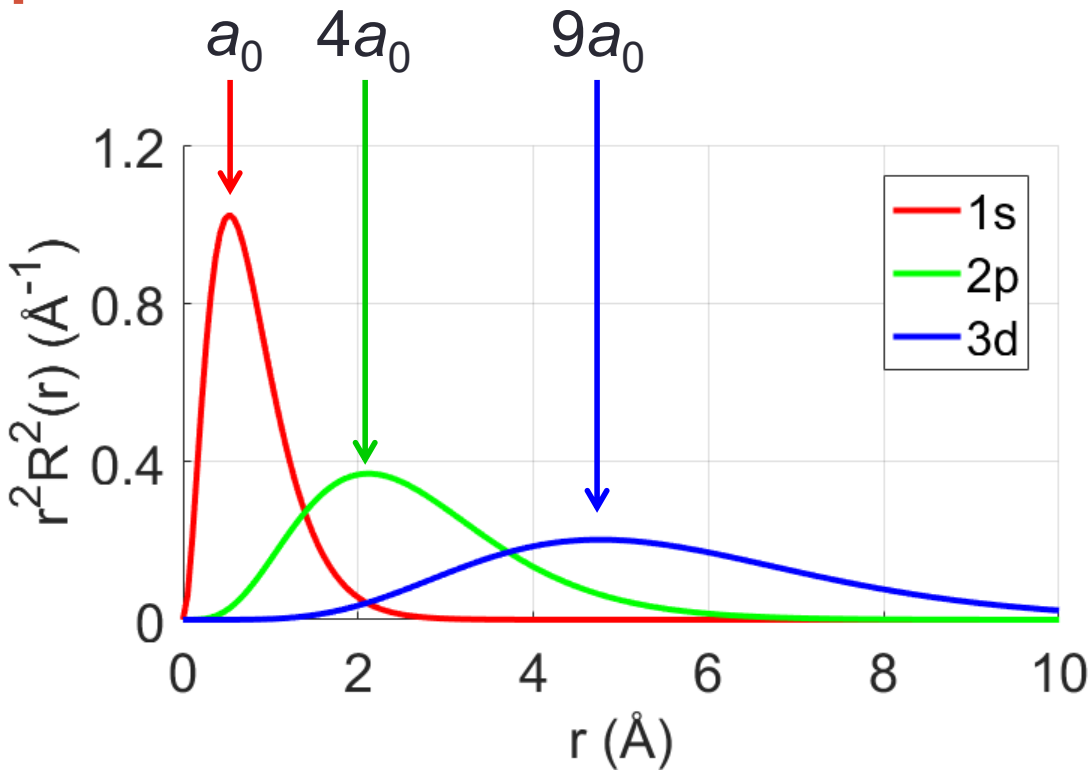
$$= -\frac{1}{4\pi\varepsilon_0}\frac{(Z-q_1-q_2)e^2}{R^2}$$

$$= -\frac{1}{4\pi\varepsilon_0}\frac{Z_{\text{eff}}e^2}{R^2}$$

$$V = -\frac{1}{4\pi\varepsilon_0}\frac{Z_{\text{eff}}e^2}{R},\ Z_{\text{eff}} = Z - q$$

Z_{eff} is effective nuclear charge.

Overlap of Orbital Shells



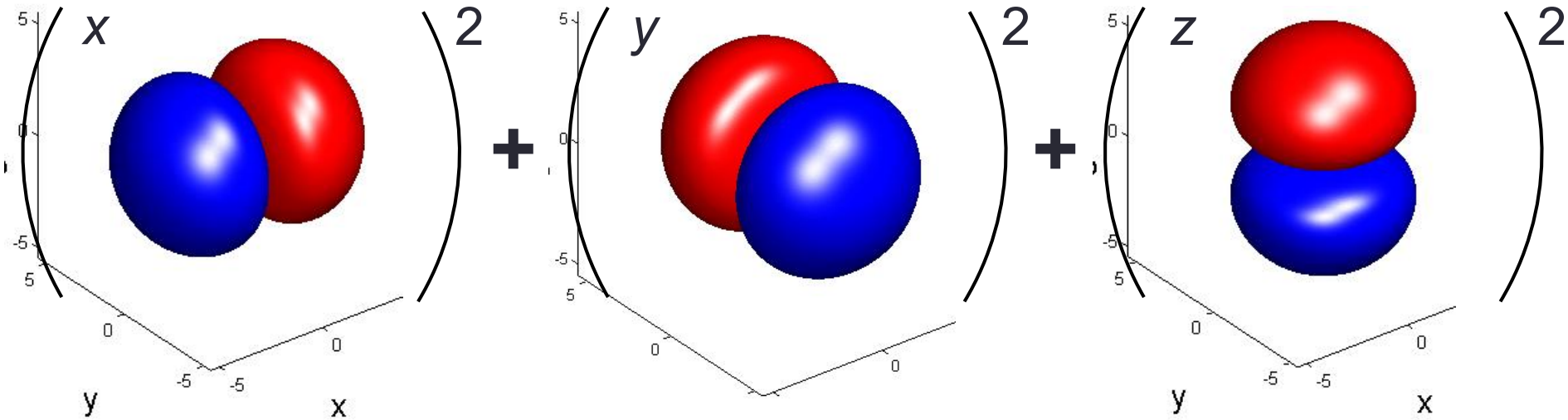
Plot $r^2R^2(r)$ vs. r

1*s*−2*p* and 2*p*−3*d*: Some overlap
1*s*−3*d*: Very little overlap

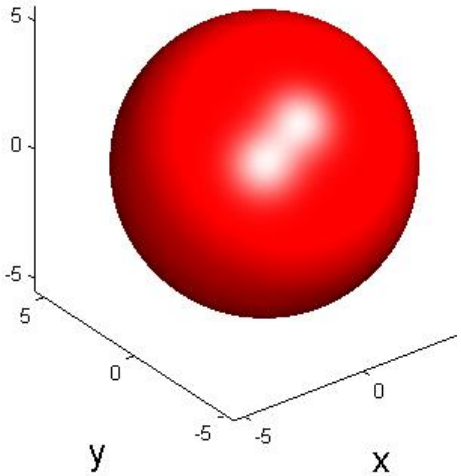
Outline

- Interactions in many-electron atoms
- Repulsion between electrons
  - Electrons in different shells
  - Electrons in the same shell
  - Electrons in the same orbital
- Aufbau principle
- Electron configurations

Overlap of *p* Orbitals

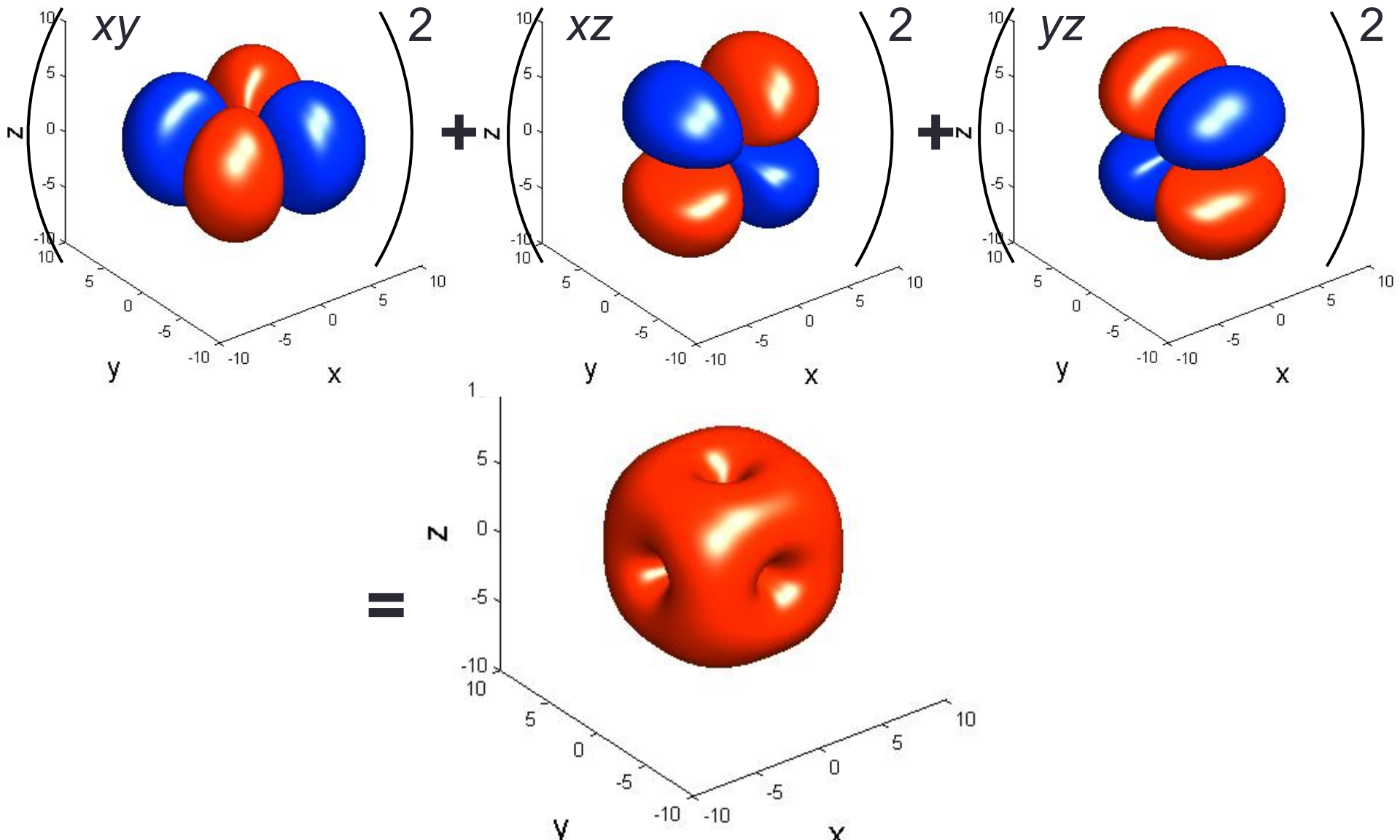


=

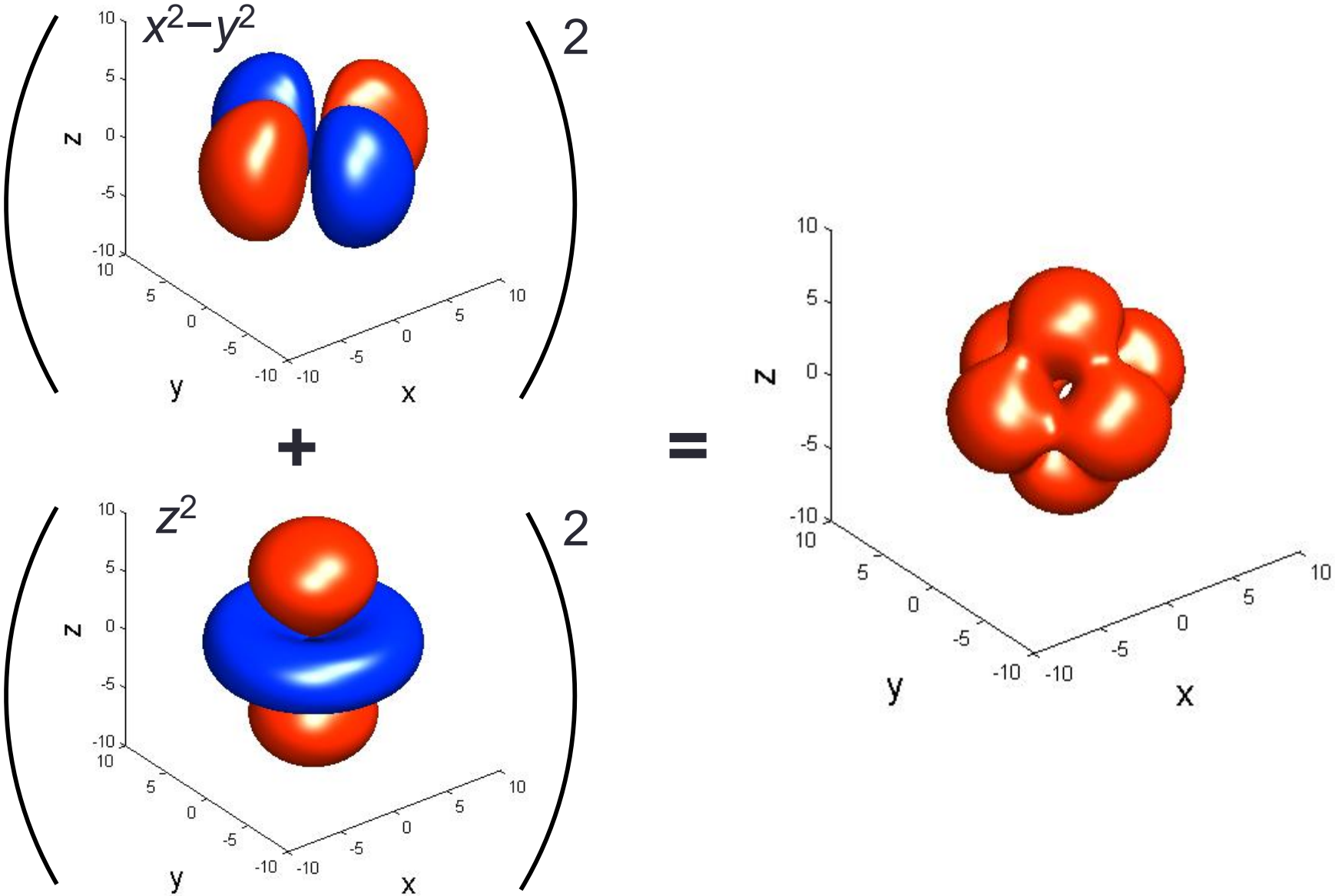


The three *p* orbitals occupy separate parts of the space.

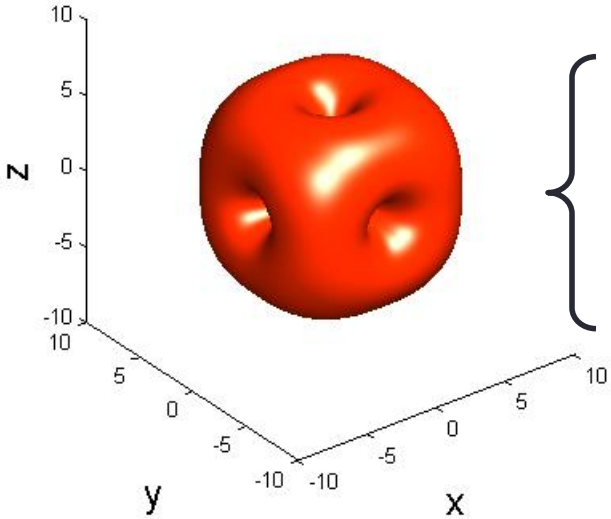
Overlap of *d* Orbitals (1)



Overlap of *d* Orbitals (2)

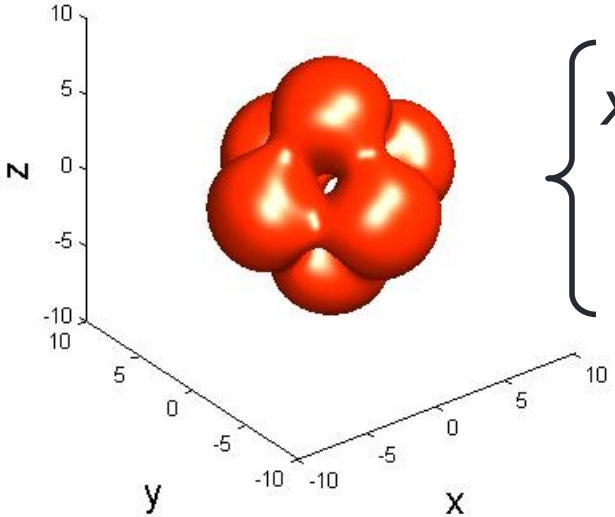


Overlap of *d* Orbitals (3)



xy
xz
yz

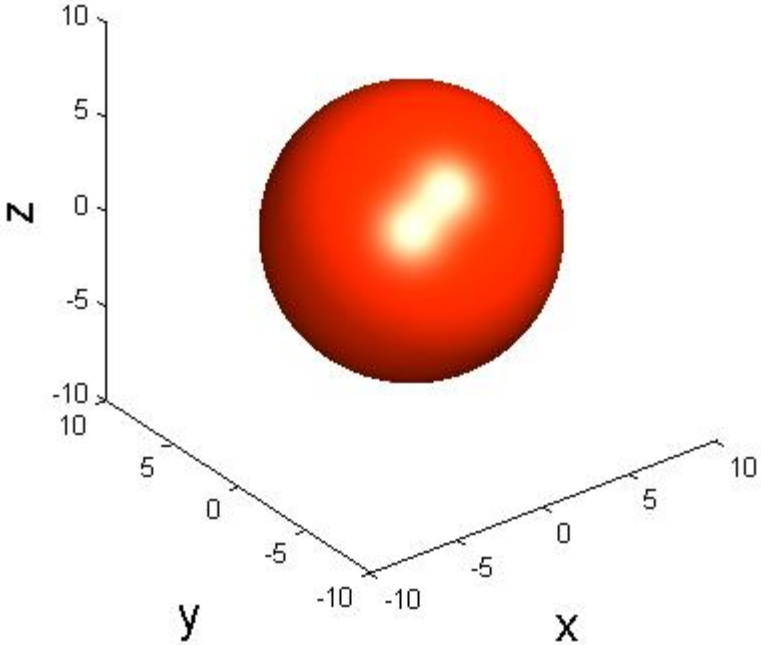
+



x^2-y^2

z^2

=

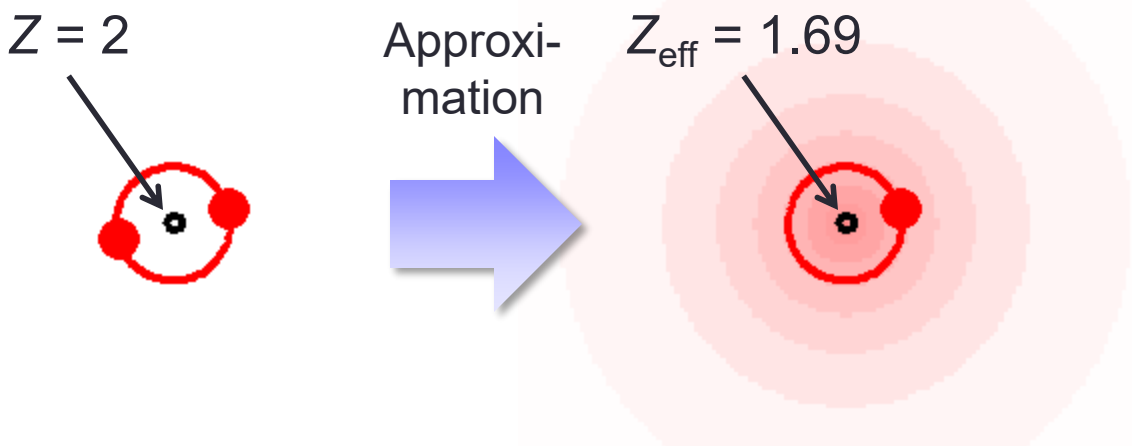


The five *d* orbitals occupy separate parts of the space.

Outline

- Interactions in many-electron atoms
- Repulsion between electrons
  - Electrons in different shells
  - Electrons in the same shell
  - Electrons in the same orbital
- Aufbau principle
- Electron configurations

Self-Consistent Field 自洽场



He: $\psi(r_1,r_2)$

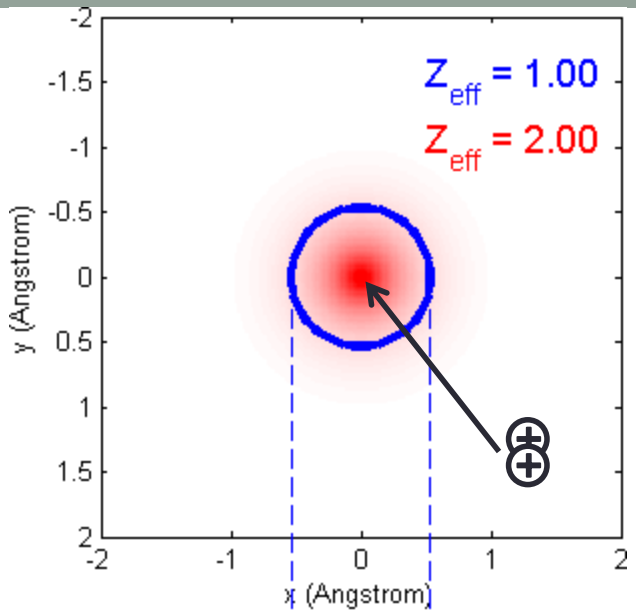
He: $\varphi_{1s}(r_1)\bullet\varphi'_{1s}(r_2)$

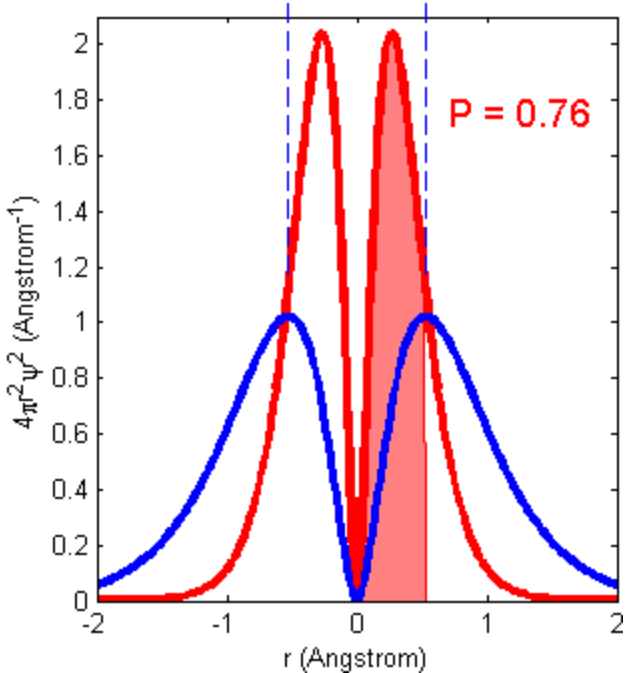
Repeated trials
until $\varphi_{1s}(r_1) = \varphi'_{1s}(r_2)$

Douglas Hartree
(Cambridge,
1897–1958)

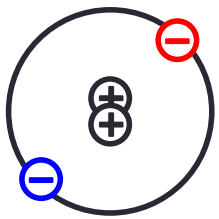
$\varphi_{1s}(r) \propto \exp(-Z_{eff}r / a_0)$, Z_{eff} is effective nuclear charge

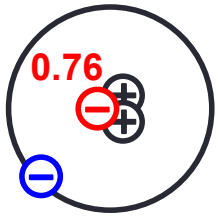
SCF Example: Helium



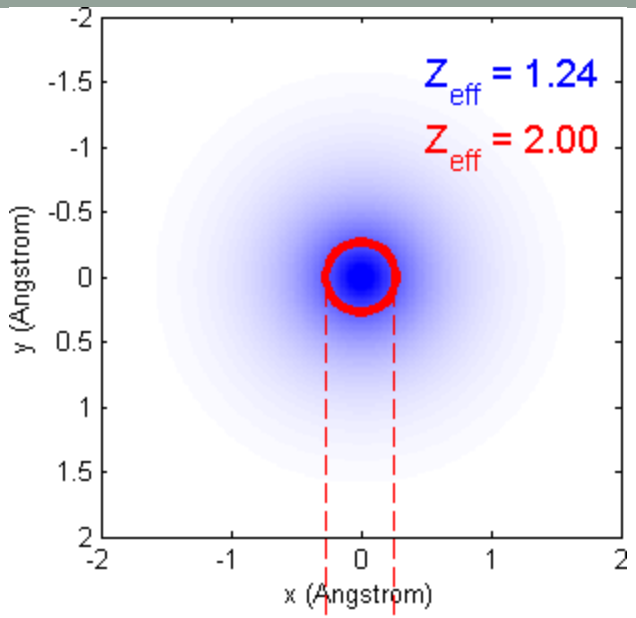


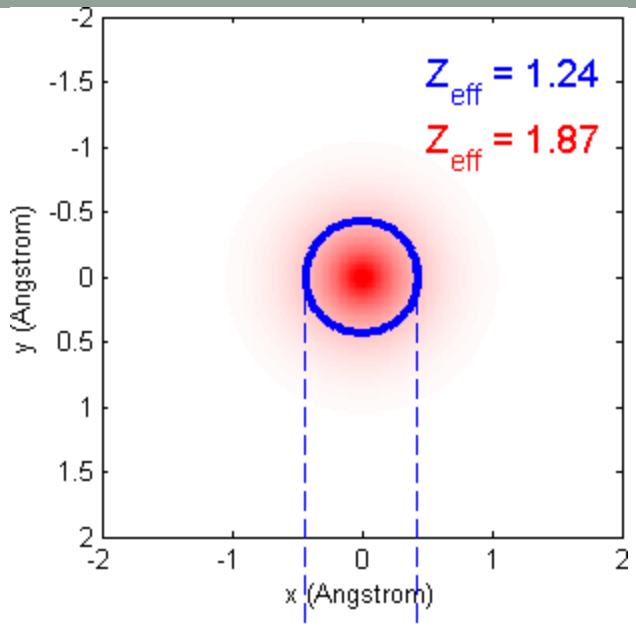
$_2$He

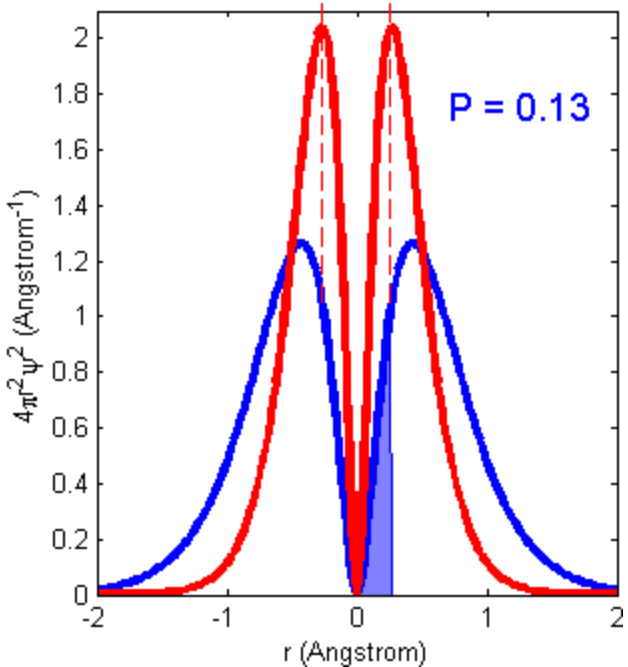


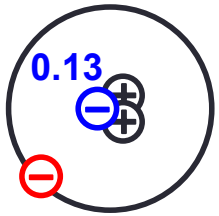


$Z_{eff} = 2 - 0.76$
$= 1.24$

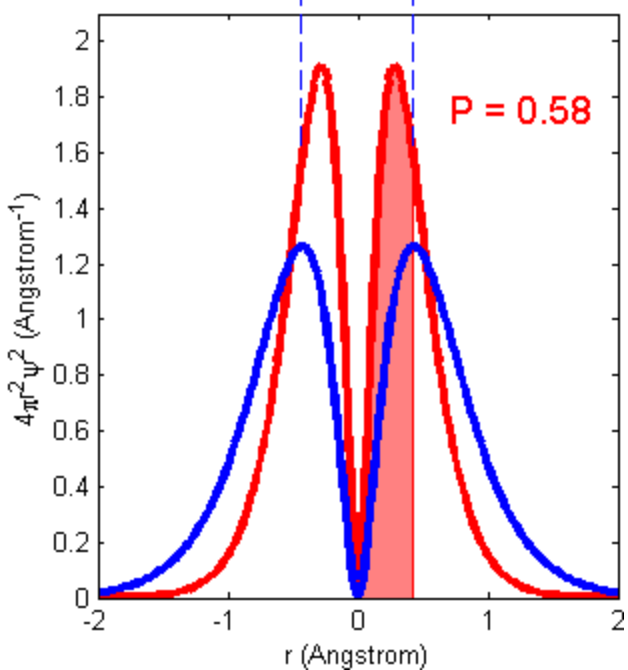


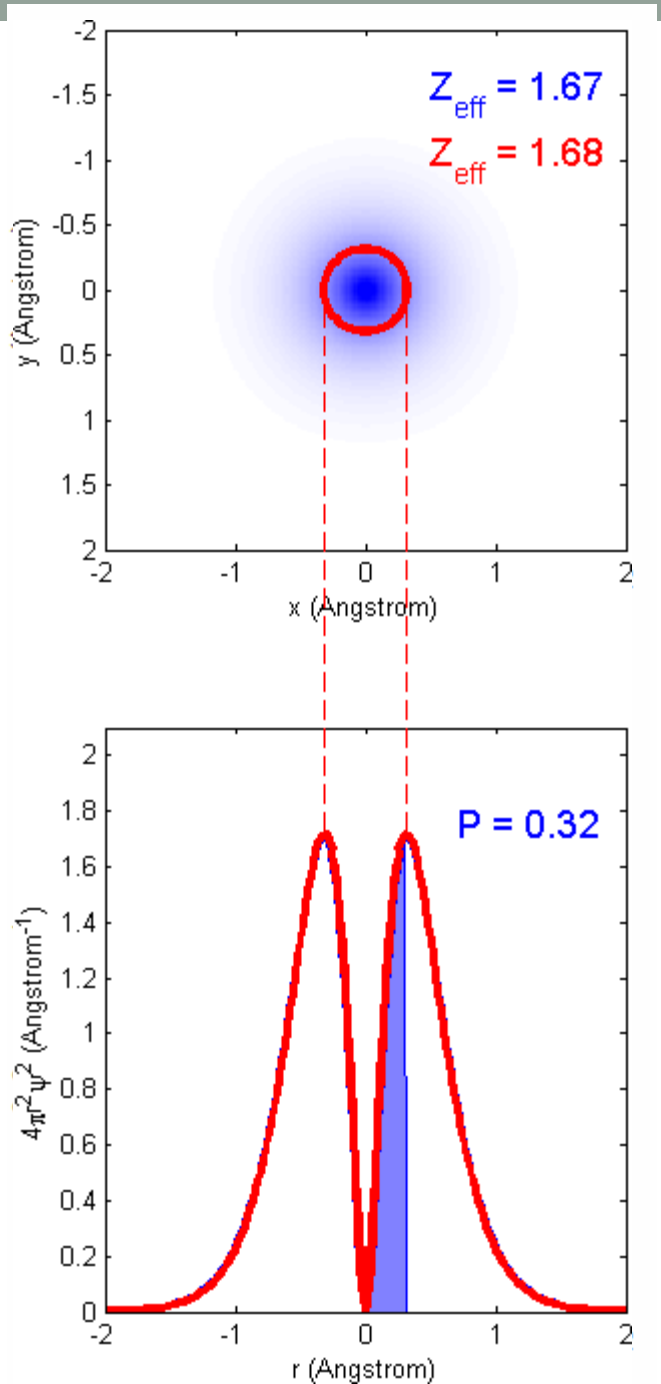


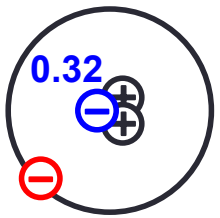




$$Z_{eff} = 2 - 0.13$$
$$= 1.87$$



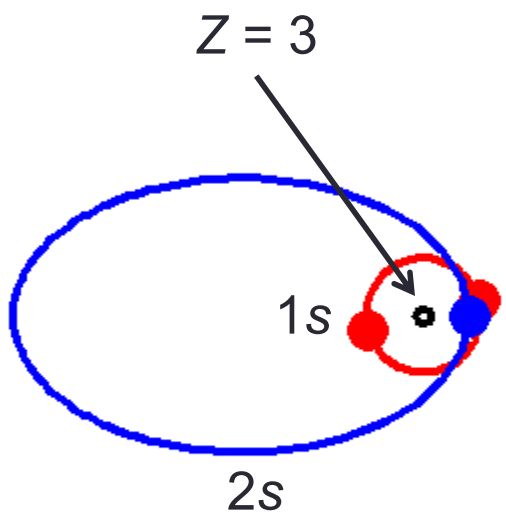


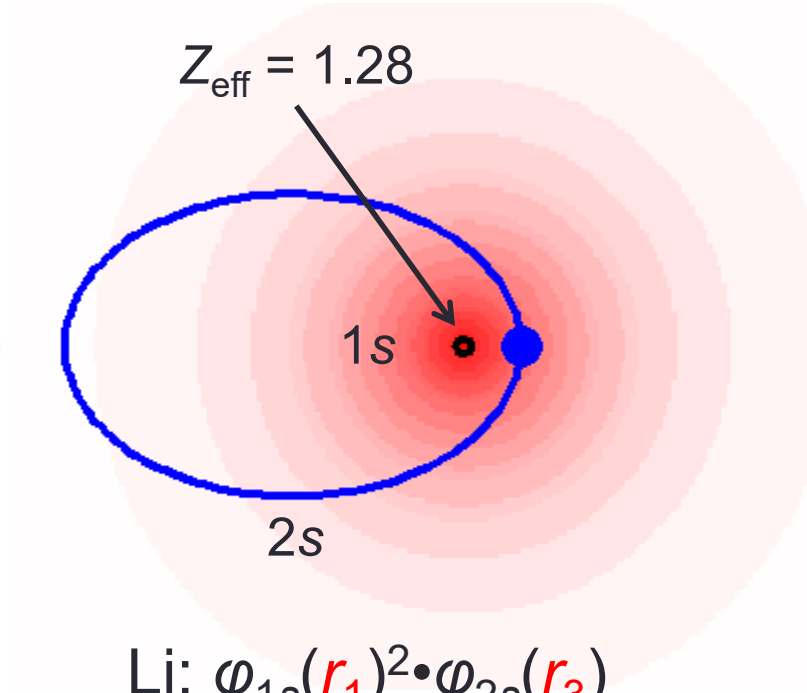


Z_{eff}(He 1s) = 2 − 0.32 = 1.68

Screening effect
(Shielding effect)
屏蔽效应

The Shell Model for Li



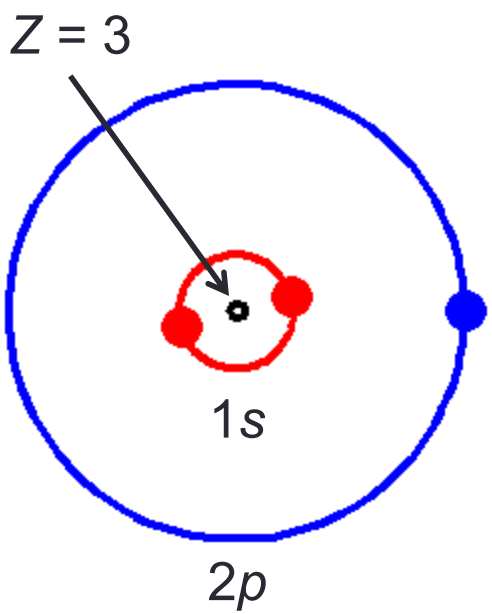


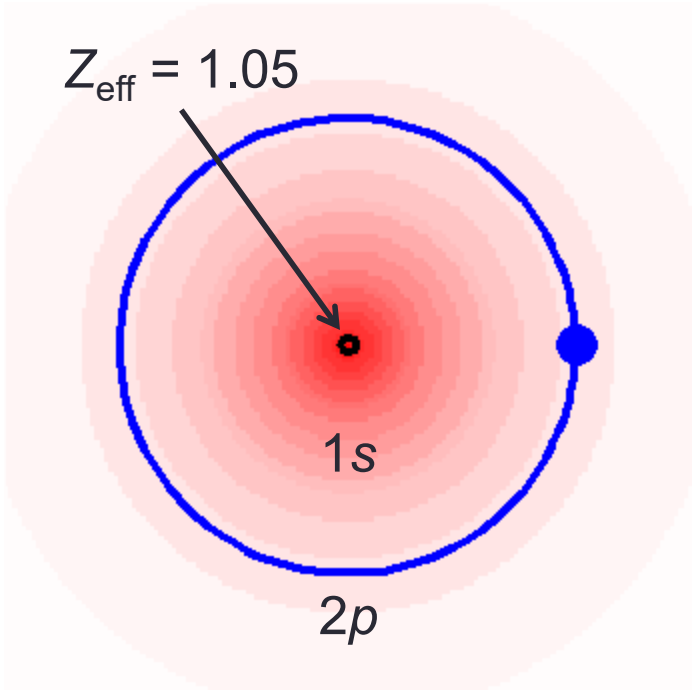
Li: $\psi(r_1,r_2,r_3)$

Li: $\varphi_{1s}(r_1)^2 \bullet \varphi_{2s}(r_3)$

$1s^2 2s^1$

The Shell Model for *Li



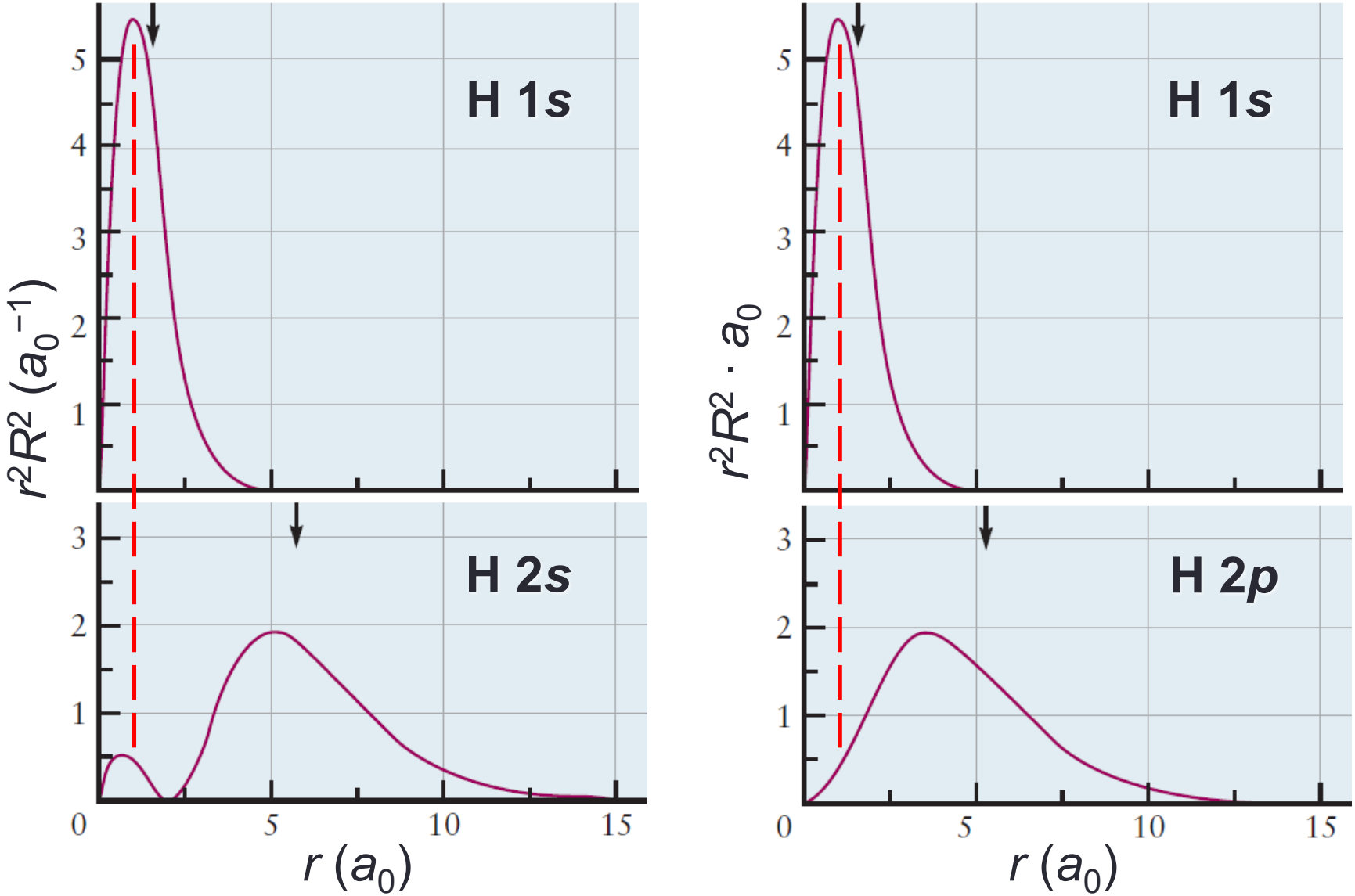


*Li: $\psi(r_1,r_2,r_3)$

*Li: $\varphi_{1s}(r_1)^2 \bullet \varphi_{2p}(r_3)$

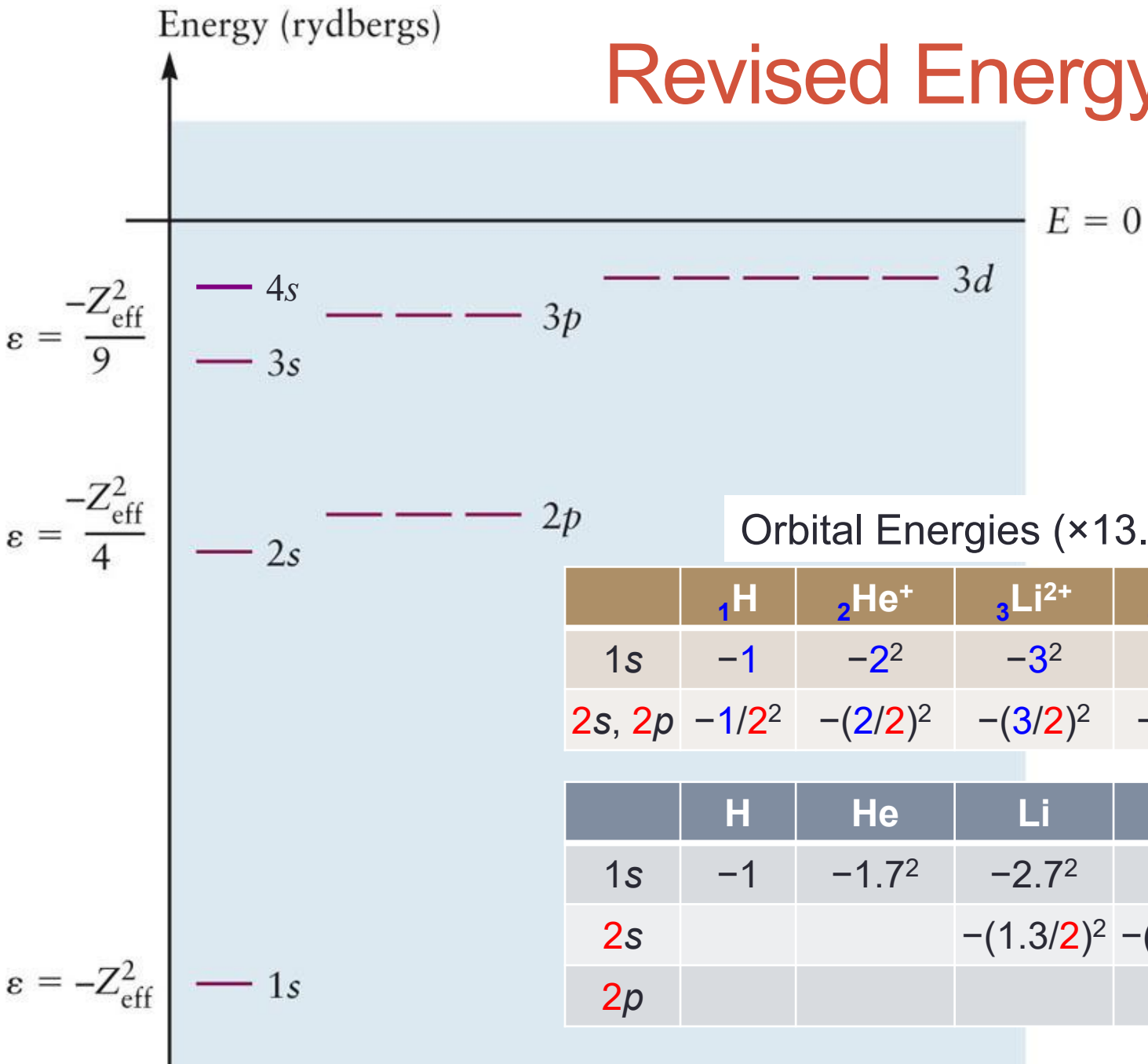
$1s^2 2p^1$

The Penetration Effect 钻穿效应



Revised Energy Levels

(OGC8 p.201)



Orbital Energies (×13.6 eV)

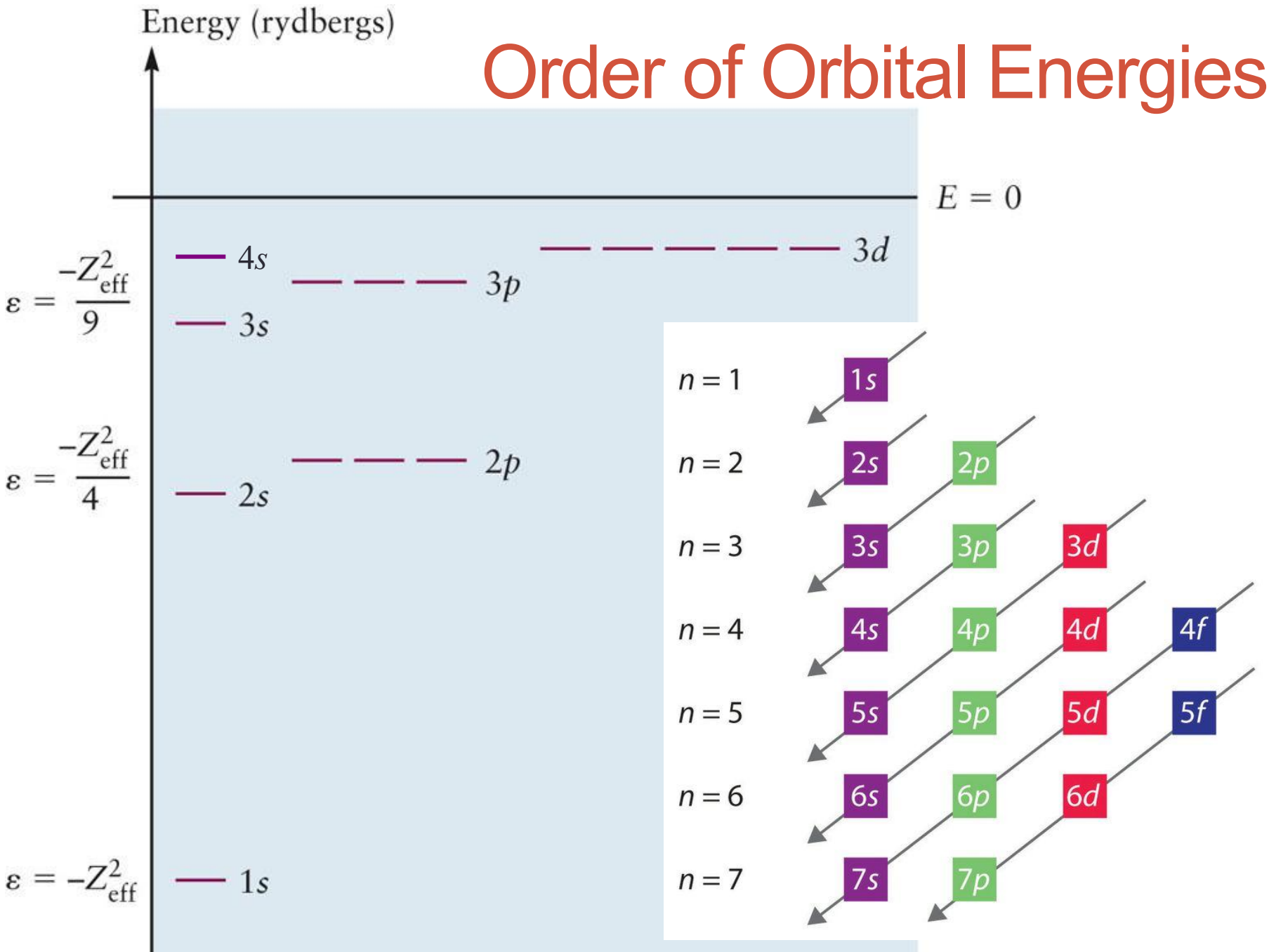
	$_1H$	$_2He^+$	$_3Li^{2+}$	$_4Be^{3+}$	$_5B^{4+}$
1*s*	-1	-2^2	-3^2	-4^2	-5^2
2*s*, 2*p*	$-1/2^2$	$-(2/2)^2$	$-(3/2)^2$	$-(4/2)^2$	$-(5/2)^2$

	H	He	Li	Be	B
1*s*	-1	-1.7^2	-2.7^2	-3.7^2	-4.7^2
2*s*			$-(1.3/2)^2$	$-(1.9/2)^2$	$-(2.6/2)^2$
2*p*					$-(2.4/2)^2$

Outline

- Interactions in many-electron atoms
- The Shell Model
  - Orbital approximation
  - Effective nuclear charge Z_{eff}
- Aufbau principle
- Electron configurations
  - Li through Ne
  - Transition metals

Order of Orbital Energies



Aufbau Principle 构造原理

Pauli 1925: Each orbital holds ≤2 electrons, one spin up ↑ and the other spin down ↓.

Hund 1925: Electrons in degenerate orbitals prefer different *m* but the same spin.

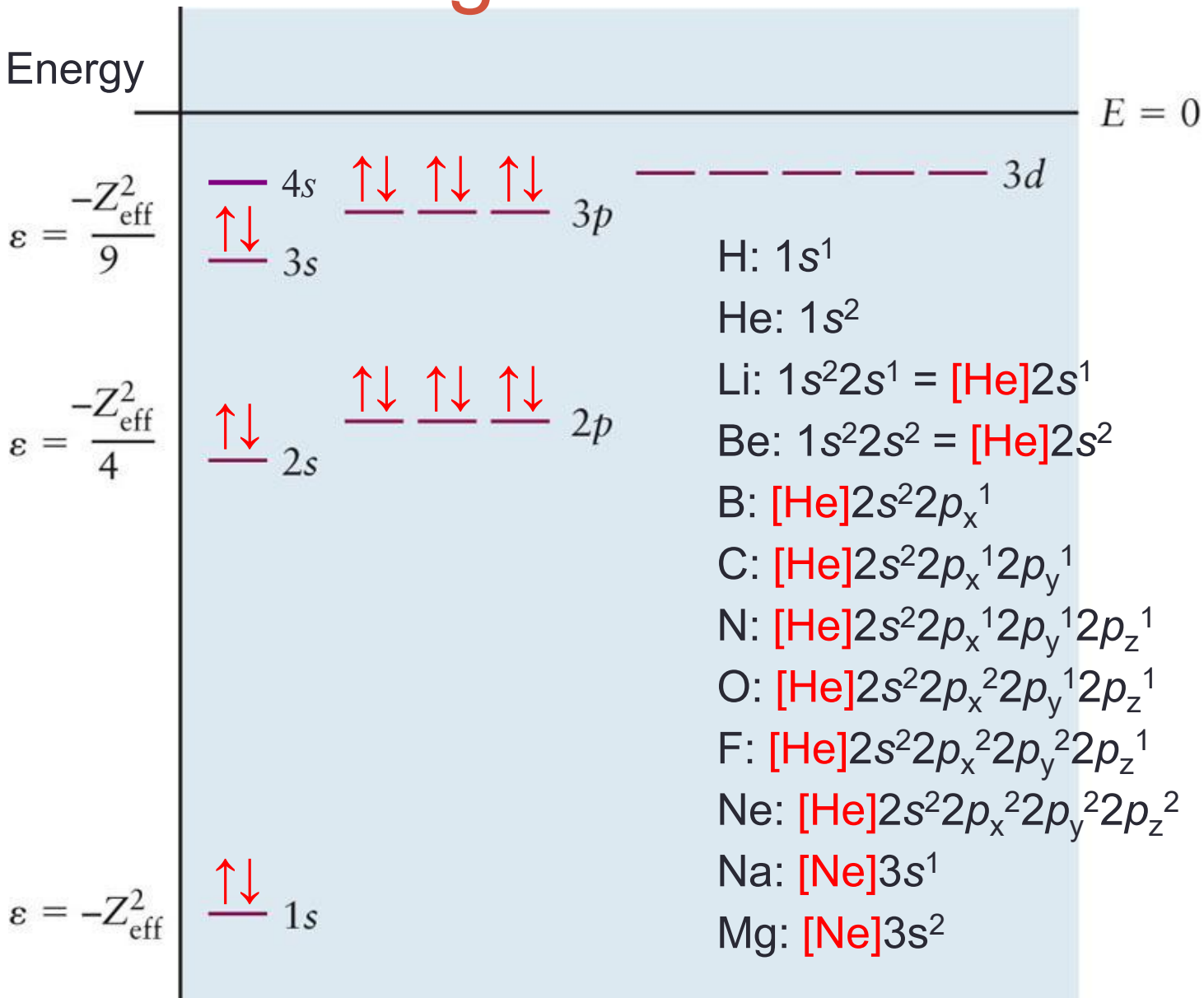
Wolfgang Pauli (1900–1958)

Friedrich Hund (1896–1997)

Outline

- Interactions in many-electron atoms
- The Shell Model
  - Orbital approximation
  - Effective nuclear charge Z_{eff}
- Aufbau principle
- Electron configurations
  - Li through Ne
  - Transition metals

Electron Configurations of H thru Ar



H: $1s^1$

He: $1s^2$

Li: $1s^2 2s^1$ = [He]$2s^1$

Be: $1s^2 2s^2$ = [He]$2s^2$

B: [He]$2s^2 2p_x^1$

C: [He]$2s^2 2p_x^1 2p_y^1$

N: [He]$2s^2 2p_x^1 2p_y^1 2p_z^1$

O: [He]$2s^2 2p_x^2 2p_y^1 2p_z^1$

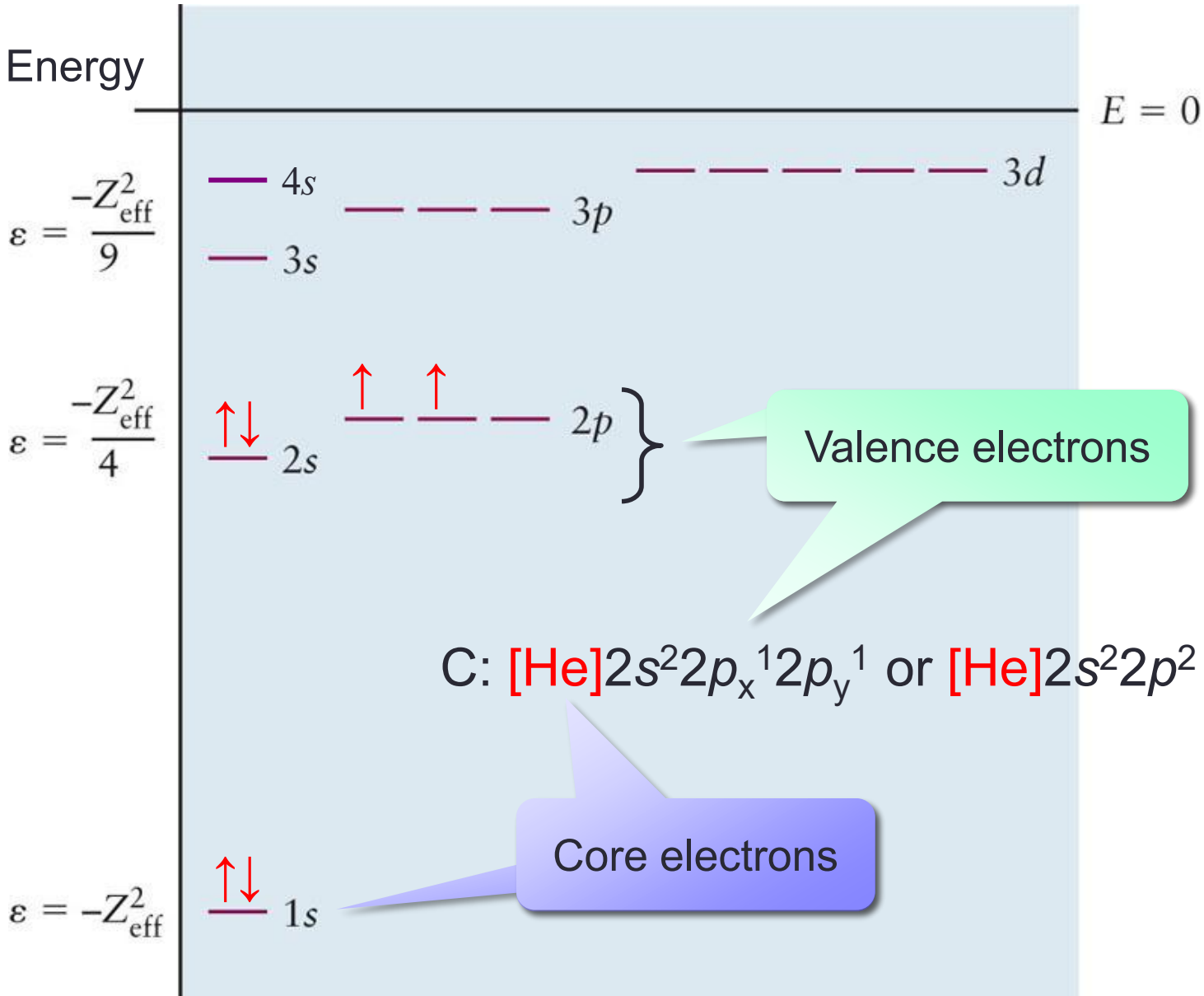
F: [He]$2s^2 2p_x^2 2p_y^2 2p_z^1$

Ne: [He]$2s^2 2p_x^2 2p_y^2 2p_z^2$

Na: [Ne]$3s^1$

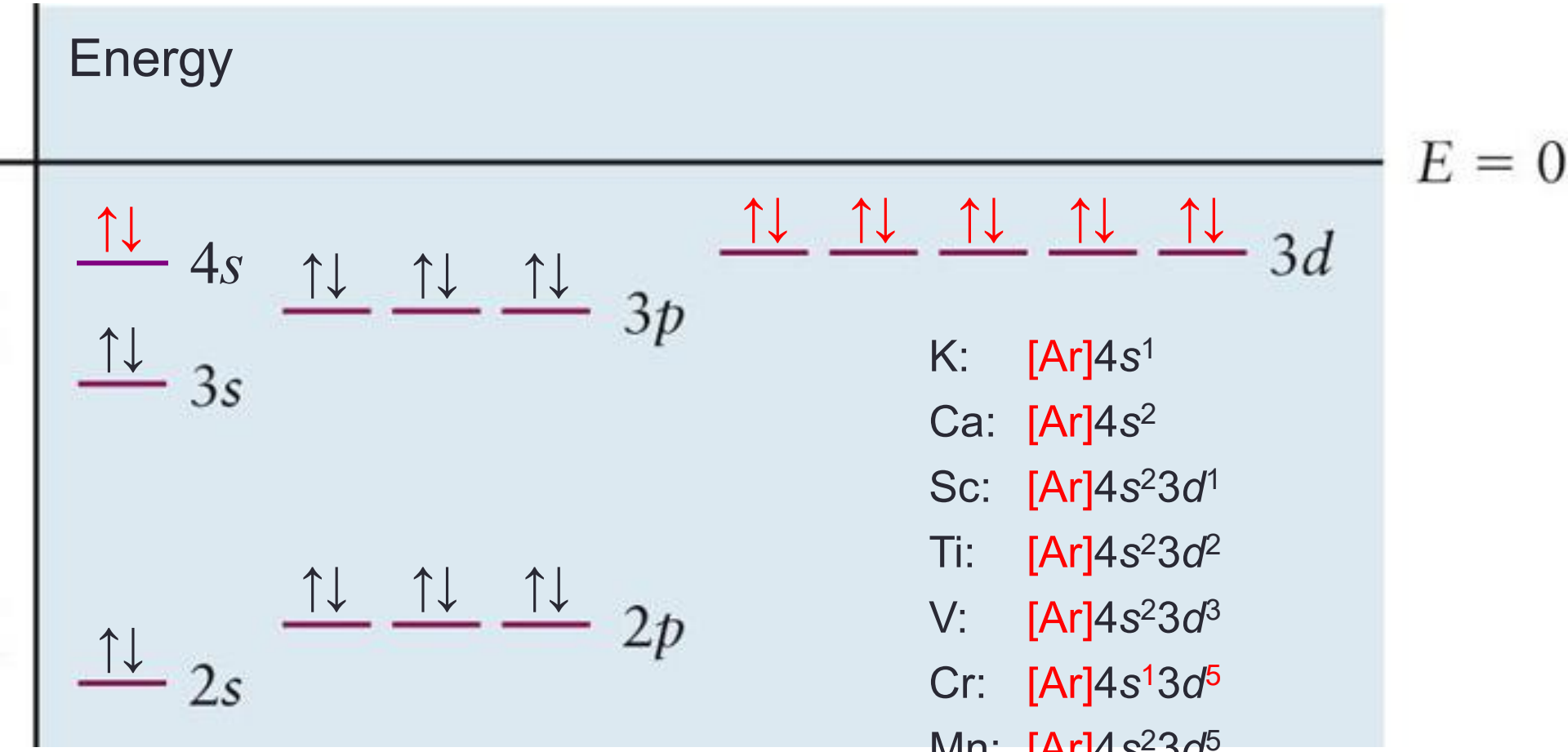
Mg: [Ne]$3s^2$

Core and Valence Electrons



C: [He]$2s^2 2p_x^1 2p_y^1$ or [He]$2s^2 2p^2$

Electron Configurations of K thru Zn



K: [Ar]$4s^1$

Ca: [Ar]$4s^2$

Sc: [Ar]$4s^2 3d^1$

Ti: [Ar]$4s^2 3d^2$

V: [Ar]$4s^2 3d^3$

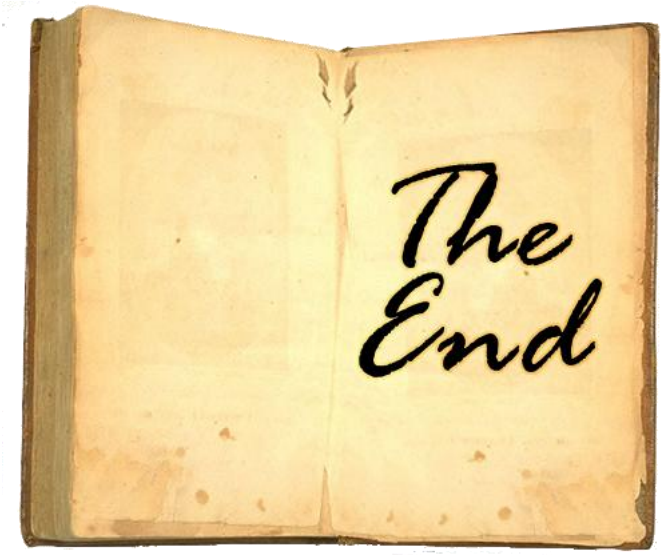
Cr: [Ar]$4s^1 3d^5$

Mn: [Ar]$4s^2 3d^5$

Fe: [Ar]$4s^2 3d^6$

Summary

- Electrons in many-electron atoms can be approximated by independent (sub)shells.
- Each subshell experiences an effective nuclear charge Z_{eff}.
- The order of the subshell energies is $1s < 2s < 2p < 3s < 3p < 4s \lesssim 3d$ …
- Aufbau principle determines how electrons fill into these subshells.

Next: Periodic Trends

- Reading:
- OGB8 §3.2, §§3.4–3.6, §5.5